

VW_BUSINESS_ANALYTICS_ADVANCED_EDA_REPORT

Overview

`VW_BUSINESS_ANALYTICS_ADVANCED_EDA_REPORT` is an executive-level analytical reporting view that consolidates business KPIs from multiple domain-specific analytical views into a single unified dataset.

The view is designed to provide a high-level exploratory data analysis (EDA) report by aggregating metrics across Sales, Customers, Operations, Inventory, and Finance. Each row represents one business metric with its calculated value, business domain, and descriptive explanation.

Purpose

The primary objective of this view is to provide decision-makers with a centralized business performance report without requiring queries against multiple analytical views.

It acts as a reusable reporting layer for dashboards, executive reporting, and business intelligence applications.

Output Structure

Column	Description
METRIC_NAME	Name of the business KPI
METRIC_VALUE	Calculated value of the metric
DOMAIN	Business domain associated with the KPI
DESCRIPTION	Business explanation of the metric

Business Domains

1. Sales & Revenue

This section summarizes revenue-related performance indicators.

Metrics

- Top Products Revenue
- Monthly Revenue
- Revenue by Category

- Average Order Value
- Promotion Impact

Business Objective

- Evaluate revenue performance
 - Identify high-performing product categories
 - Monitor customer purchasing behavior
 - Measure promotional effectiveness
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2. Customer Analytics

Provides insights into customer acquisition, retention, and lifetime value.

Metrics

- Top Customers
- New Customers Monthly
- Repeat Purchase Rate
- Customer Churn
- Customer Lifetime Value

Business Objective

- Track customer growth
 - Measure customer loyalty
 - Identify customer churn
 - Estimate long-term customer value
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3. Orders & Fulfillment

Measures operational efficiency of the order lifecycle.

Metrics

- Order Cancellations
- Late Deliveries
- Payment Mismatches
- Return Reasons

Business Objective

- Improve delivery performance
 - Reduce operational issues
 - Monitor returns
 - Identify payment inconsistencies
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4. Inventory & Suppliers

Evaluates inventory utilization and supplier efficiency.

Metrics

- Out of Stock Products
- High Margin Products
- Stock Turnover Ratio
- Supplier Performance

Business Objective

- Improve inventory planning
 - Identify profitable products
 - Monitor supplier delivery performance
 - Optimize inventory turnover
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5. Finance & Risk

Monitors financial performance and operational risk indicators.

Metrics

- Payment Success Rate
- Currency Exchange Impact
- Fraud Flags
- Logistics Cost
- Campaign ROI
- Refund Processing Time
- Compliance Issues

Business Objective

- Monitor payment reliability
 - Evaluate foreign exchange impact
 - Detect potential fraud
 - Measure logistics expenses
 - Evaluate marketing effectiveness
 - Track refund efficiency
 - Monitor regulatory compliance
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Source Analytics Views

The report aggregates data from the following analytical views:

Sales & Revenue

- VW_TOP_PRODUCTS_ANALYTICS
- VW_MONTHLY_REVENUE_ANALYTICS
- VW_REVENUE_BY_CATEGORY_ANALYTICS
- VW_AVG_ORDER_VALUE_ANALYTICS
- VW_PROMOTION_IMPACT_ANALYTICS

Customer Analytics

- VW_TOP_CUSTOMERS_ANALYTICS
- VW_NEW_CUSTOMERS_MONTHLY_ANALYTICS
- VW_REPEAT_PURCHASE_RATE_ANALYTICS
- VW_CUSTOMER_CHURN_ANALYTICS
- VW_CUSTOMER_LIFETIME_VALUE_ANALYTICS

Orders & Fulfillment

- VW_ORDER_CANCELLATION_ANALYTICS
- VW_LATE_DELIVERIES_ANALYTICS
- VW_PAYMENT_MISMATCH_ANALYTICS
- VW_RETURN_REASON_ANALYTICS

Inventory & Suppliers

- VW_OUT_OF_STOCK_PRODUCTS_ANALYTICS
- VW_HIGH_MARGIN_PRODUCTS_ANALYTICS
- VW_STOCK_TURNOVER_ANALYTICS
- VW_SUPPLIER_PERFORMANCE_ANALYTICS

Finance & Risk

- VW_PAYMENT_SUCCESS_RATE_ANALYTICS
- VW_CURRENCY_EXCHANGE_IMPACT_ANALYTICS
- VW_FRAUD_DETECTION_ANALYTICS
- VW_LOGISTICS_COST_ANALYTICS
- VW_CAMPAIGN_ROI_ANALYTICS
- VW_REFUND_PROCESSING_TIME_ANALYTICS
- VW_COMPLIANCE_CHECKS_ANALYTICS

Design Characteristics

- Centralized executive KPI reporting
 - Domain-wise organization of business metrics
 - Built using `UNION ALL`
 - Aggregates KPIs from specialized analytical views
 - One metric per row for simplified reporting
 - Reusable data source for dashboards and BI tools
 - Reduces the need to query multiple analytical views individually
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Typical Use Cases

- Executive dashboards
 - Exploratory Data Analysis (EDA)
 - Business performance monitoring
 - KPI scorecards
 - Monthly and quarterly reporting
 - Operational review meetings
 - Financial performance analysis
 - Power BI and Tableau dashboards
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Benefits

- Consolidates multiple business metrics into a single view
 - Provides consistent KPI definitions across reports
 - Simplifies dashboard development
 - Improves maintainability by centralizing business logic
 - Supports reusable analytics for downstream reporting
 - Enables quick assessment of overall business health
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Conclusion

`VW_BUSINESS_ANALYTICS_ADVANCED_EDA_REPORT` serves as a unified executive EDA reporting layer by combining key performance indicators from multiple analytical domains into a single business-oriented view. It provides stakeholders with a comprehensive summary of organizational performance, enabling faster decision-making, streamlined reporting, and efficient integration with business intelligence and visualization platforms.